Year 8

REVISION PACK

Unit 4

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 4 · Area of a triangle

Q	Working	Answer
the area of a triangle		
1.1	$\frac{1}{2} \times 7 \times 4 = 14$	**14 cm²**
1.2	$\frac{1}{2} \times 10 \times 3 = 15$	**15 cm²**
1.3	$\frac{1}{2} \times 9 \times 6 = 27$	**27 cm²**
1.4	$\frac{1}{2} \times 5 \times 8 = 20$	**20 cm²**
1.5	$\frac{1}{2} \times 12 \times 7 = 42$	**42 cm²**
1.6	$\frac{1}{2} \times 11 \times 4 = 22$	**22 cm²**

Unit 4 · Area of a trapezium

Q	Working	Answer
the area of a trapezium		
2.1	$(6 + 4) \div 2 = 5$ $5 \times 8 = 40$	**40 cm²**
2.2	$(5 + 11) \div 2 = 8$ $8 \times 6 = 48$	**48 cm²**
2.3	$(3 + 9) \div 2 = 6$ $6 \times 7 = 42$	**42 cm²**
2.4	$(8 + 12) \div 2 = 10$ $10 \times 5 = 50$	**50 cm²**
2.5	$(2 + 10) \div 2 = 6$ $6 \times 9 = 54$	**54 cm²**

2.6	$(7+13)\div 2=10$ $10\times 4=40$	**40 cm²**

Unit 4 · Volume and surface area

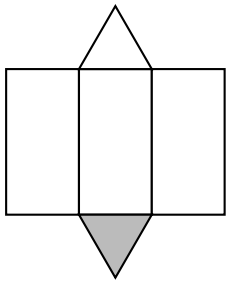
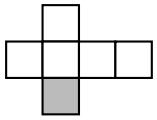
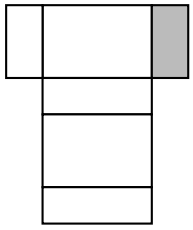
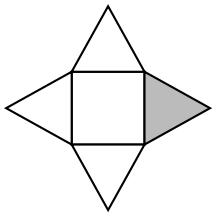
Q	Working	Answer
volume and surface area of a cuboid		
3.1	**(a)** $5\times 6\times 4=120$ **(b)** $5\times 6=30,\ 5\times 4=20,\ 6\times 4=24$ $2\times(30+20+24)=148$	**(a) 120 cm³** **(b) 148 cm²**
3.2	**(a)** $9\times 3\times 2=54$ **(b)** $9\times 3=27,\ 9\times 2=18,\ 3\times 2=6$ $2\times(27+18+6)=102$	**(a) 54 cm³** **(b) 102 cm²**
3.3	**(a)** $7\times 7\times 5=245$ **(b)** $7\times 7=49,\ 7\times 5=35,\ 7\times 5=35$ $2\times(49+35+35)=238$	**(a) 245 cm³** **(b) 238 cm²**
3.4	**(a)** $10\times 6\times 3=180$ **(b)** $10\times 6=60,\ 10\times 3=30,\ 6\times 3=18$ $2\times(60+30+18)=216$	**(a) 180 cm³** **(b) 216 cm²**
3.5	**(a)** $8\times 5\times 4=160$ **(b)** $8\times 5=40,\ 8\times 4=32,\ 5\times 4=20$ $2\times(40+32+20)=184$	**(a) 160 cm³** **(b) 184 cm²**
3.6	**(a)** $11\times 4\times 2=88$ **(b)** $11\times 4=44,\ 11\times 2=22,\ 4\times 2=8$ $2\times(44+22+8)=148$	**(a) 88 cm³** **(b) 148 cm²**

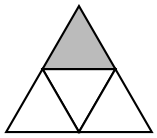
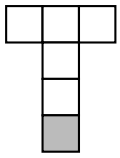
Unit 4 · Capacity and units

Q	Working	Answer
litres and units of volume		

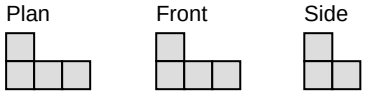
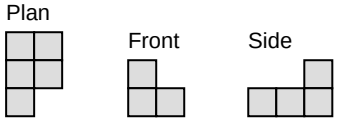
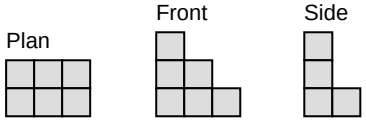
4.1	**(a)** $45 \times 30 \times 20 = 27\,000\ \text{cm}^3$ $27\,000 \div 1000 = 27$ **(b)** $50 \times 1000 = 50\,000$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $4.6 \times 1000 = 4600$	**(a) 27 litres** **(b) 50 000 mm³** **(c) shown**
4.2	**(a)** $60 \times 30 \times 25 = 45\,000\ \text{cm}^3$ $45\,000 \div 1000 = 45$ **(b)** $8 \times 1000 = 8\,000$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $3.2 \times 1000 = 3200$	**(a) 45 litres** **(b) 8 000 mm³** **(c) shown**
4.3	**(a)** $50 \times 40 \times 15 = 30\,000\ \text{cm}^3$ $30\,000 \div 1000 = 30$ **(b)** $12 \times 1000 = 12\,000$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $1.75 \times 1000 = 1750$	**(a) 30 litres** **(b) 12 000 mm³** **(c) shown**
4.4	**(a)** $80 \times 25 \times 30 = 60\,000\ \text{cm}^3$ $60\,000 \div 1000 = 60$ **(b)** $4.5 \times 1000 = 4\,500.0$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $6.8 \times 1000 = 6800$	**(a) 60 litres** **(b) 4 500.0 mm³** **(c) shown**
4.5	**(a)** $35 \times 20 \times 40 = 28\,000\ \text{cm}^3$ $28\,000 \div 1000 = 28$ **(b)** $30 \times 1000 = 30\,000$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $0.9 \times 1000 = 900$	**(a) 28 litres** **(b) 30 000 mm³** **(c) shown**
4.6	**(a)** $70 \times 30 \times 20 = 42\,000\ \text{cm}^3$ $42\,000 \div 1000 = 42$ **(b)** $2.6 \times 1000 = 2\,600.0$ **(c)** $1\ \text{m}^3 = 1000\ \text{litres}$ $4.25 \times 1000 = 4250$	**(a) 42 litres** **(b) 2 600.0 mm³** **(c) shown**

Unit 4 · Nets

Q	Working	Answer
completing a net and naming the solid		
5.1	**(a)** **(b)** **(c)** 	**(a) a triangle on the other end of the middle rectangle** **(b) Triangular prism** **(c) A sketch of a triangular prism**
5.2	**(a)** **(b)** **(c)** 	**(a) a square on any free edge (e.g. below the second square)** **(b) Cube** **(c) A sketch of a cube**
5.3	**(a)** **(b)** **(c)** 	**(a) a 1 by 2 rectangle on the right of the top face** **(b) Cuboid** **(c) A sketch of a cuboid**
5.4	**(a)** **(b)** **(c)** 	**(a) a triangle on the fourth side of the square** **(b) Square-based pyramid** **(c) A sketch of a square-based pyramid**

5.5	**(a)** **(b)** **(c)** 	**(a) a triangle on the free edge of the middle triangle** **(b) Triangular-based pyramid (tetrahedron)** **(c) A sketch of a triangular-based pyramid**
5.6	**(a)** **(b)** **(c)** 	**(a) a square on any free edge (e.g. at the bottom)** **(b) Cube** **(c) A sketch of a cube**

Unit 4 · Plans and elevations

Q	Working	Answer
drawing plans and elevations		
6.1	plan: look from above front heights: $2, 1, 1$ side heights: $2, 1$ 	**See the drawings.**
6.2	plan: look from above front heights: $2, 1$ side heights: $1, 1, 2$ 	**See the drawings.**
6.3	plan: look from above front heights: $3, 2, 1$ side heights: $3, 1$ 	**See the drawings.**

6.4	plan: look from above front heights: $1, 2, 1$ side heights: $2, 1$ Plan  Front  Side 	**See the drawings.**
6.5	plan: look from above front heights: $2, 2$ side heights: $1, 1, 2$ Plan  Front  Side 	**See the drawings.**
6.6	plan: look from above front heights: $1, 1, 1$ side heights: $1, 1$ Plan  Front  Side 	**See the drawings.**

Unit 4 · Volume and surface area

Q	Working	Answer
a block with a hole		
7.1	**(a)** $16 \times 12 \times 20 = 3840$ $4 \times 4 \times 20 = 320$ $3840 - 320 = 3520$ **(b)** front and back : $2 \times (192 - 16) = 352$ top and bottom : $2 \times 16 \times 20 = 640$ two sides : $2 \times 12 \times 20 = 480$ inside the hole : $4 \times 4 \times 20 = 320$ total $= 1792$	**(a) 3520 cm³** **(b) 1792 cm²**

7.2	**(a)** $12 \times 10 \times 25 = 3000$ $4 \times 4 \times 25 = 400$ $3000 - 400 = 2600$ **(b)** front and back : $2 \times (120 - 16) = 208$ top and bottom : $2 \times 12 \times 25 = 600$ two sides : $2 \times 10 \times 25 = 500$ inside the hole : $4 \times 4 \times 25 = 400$ total $= 1708$	**(a) 2600 cm³** **(b) 1708 cm²**
7.3	**(a)** $15 \times 9 \times 30 = 4050$ $3 \times 3 \times 30 = 270$ $4050 - 270 = 3780$ **(b)** front and back : $2 \times (135 - 9) = 252$ top and bottom : $2 \times 15 \times 30 = 900$ two sides : $2 \times 9 \times 30 = 540$ inside the hole : $4 \times 3 \times 30 = 360$ total $= 2052$	**(a) 3780 cm³** **(b) 2052 cm²**
7.4	**(a)** $20 \times 14 \times 10 = 2800$ $6 \times 6 \times 10 = 360$ $2800 - 360 = 2440$ **(b)** front and back : $2 \times (280 - 36) = 488$ top and bottom : $2 \times 20 \times 10 = 400$ two sides : $2 \times 14 \times 10 = 280$ inside the hole : $4 \times 6 \times 10 = 240$ total $= 1408$	**(a) 2440 cm³** **(b) 1408 cm²**
7.5	**(a)** $16 \times 16 \times 12 = 3072$ $8 \times 8 \times 12 = 768$ $3072 - 768 = 2304$ **(b)** front and back : $2 \times (256 - 64) = 384$ top and bottom : $2 \times 16 \times 12 = 384$ two sides : $2 \times 16 \times 12 = 384$ inside the hole : $4 \times 8 \times 12 = 384$ total $= 1536$	**(a) 2304 cm³** **(b) 1536 cm²**

7.6	**(a)** $10 \times 12 \times 40 = 4800$ $2 \times 2 \times 40 = 160$ $4800 - 160 = 4640$ **(b)** front and back : $2 \times (120 - 4) = 232$ top and bottom : $2 \times 10 \times 40 = 800$ two sides : $2 \times 12 \times 40 = 960$ inside the hole : $4 \times 2 \times 40 = 320$ total $= 2312$	**(a) 4640 cm³** **(b) 2312 cm²**

Unit 4 · Area problems

Q	Working	Answer
the area of a path		
8.1	$50 \text{ cm} = 0.5 \text{ m}$ big $= 10 \times 10 = 100$ lawn $= 9 \times 9 = 81$ $100 - 81 = 19$	**19 m²**
8.2	$100 \text{ cm} = 1 \text{ m}$ big $= 9 \times 9 = 81$ lawn $= 7 \times 7 = 49$ $81 - 49 = 32$	**32 m²**
8.3	$50 \text{ cm} = 0.5 \text{ m}$ big $= 11 \times 7 = 77$ lawn $= 10 \times 6 = 60$ $77 - 60 = 17$	**17 m²**
8.4	$100 \text{ cm} = 1 \text{ m}$ big $= 17 \times 17 = 289$ lawn $= 15 \times 15 = 225$ $289 - 225 = 64$	**64 m²**
8.5	$50 \text{ cm} = 0.5 \text{ m}$ big $= 6 \times 13 = 78$ lawn $= 5 \times 12 = 60$ $78 - 60 = 18$	**18 m²**

8.6	$25\text{ cm} = 0.25\text{ m}$ $\text{big} = 3.5 \times 3.5 = 12.25$ $\text{lawn} = 3 \times 3 = 9$ $12.25 - 9 = 3.25$	**3.25 m²**

Unit 4 · Area problems

Q	Working	Answer
a parallelogram and a triangle with the same area		
9.1	$\text{parallelogram} = 12 \times 5 = 60$ $\frac{1}{2} \times b \times 8 = 60$ $4b = 60$ $b = 15$	**shown**
9.2	$\text{parallelogram} = 15 \times 4 = 60$ $\frac{1}{2} \times b \times 10 = 60$ $5b = 60$ $b = 12$	**shown**
9.3	$\text{parallelogram} = 7 \times 6 = 42$ $\frac{1}{2} \times b \times 12 = 42$ $6b = 42$ $b = 7$	**shown**
9.4	$\text{parallelogram} = 10 \times 9 = 90$ $\frac{1}{2} \times b \times 15 = 90$ $7.5b = 90$ $b = 12$	**shown**
9.5	$\text{parallelogram} = 16 \times 3 = 48$ $\frac{1}{2} \times b \times 6 = 48$ $3b = 48$ $b = 16$	**shown**

9.6	$\text{parallelogram} = 9 \times 8 = 72$ $\frac{1}{2} \times b \times 16 = 72$ $8b = 72$ $b = 9$	**shown**